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On the applicability of oil shale to palaeomagnetic investigations

M. Krs [a], M. Krsová [a], L. Kouklíková [a], P. Pruner [a] and F. Valín [b]

[a] *Geofyzika S.C., Geologická 2, 152 00 Prague 5 — Barrandov, Czech and Slovak Federal Rep.*
[b] *Central Geological Survey, Malostranské nám. 19, 118 21 Prague 1 — Malá Strana, Czech and Slovak Federal Rep.*

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ABSTRACT

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Samples of oil shales of Autunian age from the Krkonoše (Riesengebirge) Piedmont basin, northern Bohemia, which represent rocks with fossil micro-organic matter in the initial stage of carbonification, were subjected to palaeomagnetic investigations. The palaeomagnetic data obtained from these rocks are fully in accordance with those derived previously on rocks of the red-bed facies in the same area. Preliminary results obtained on Lower Carboniferous coal-bearing shales collected from the Upper Silesian basin, Ostrava–Karviná region, indicate also that these rocks, characterized by a higher degree of carbonification, are suitable for palaeomagnetic investigations. These results with those of laboratory carbonification indicate that such rocks as bituminous rocks, black shales, oil shales, coal-bearing shales or generally rocks containing micro-organic matter at different stages of natural carbonification are worthy of being studied palaeomagnetically.

1. Introduction

It has been found that greigite or greigite–smythite mineralization is mineralogically extremely unstable. As shown by laboratory investigations, depending on thermal history this mineralization can be the source of other minerals, such as pyrrhotite, pyrite, marcasite, γ-, η-, α-Fe_2O_3 or magnetite (Krs et al., 1992). So far, the existence of this mineralization has been proven in the Bohemian Massif solely in connection with Miocene rocks containing fossil micro-organic matter. Greigite or greigite–smythite mineralization is preserved first of all in young rocks. According to the results of laboratory analyses, rocks of similar facies, but partly or completely carbonified, should contain products of phase or mineralogical changes of the original greigite or greigite–smythite mineralization. For laboratory research, therefore, we collected samples of oil shales of Autunian age, which are rocks with fossil micro-organic matter in the initial stage of the carbonification process. Research was completed with studies of samples of the lower Carboniferous that were already fully carbonified. In this way, the results of laboratory-carbonified rocks with original greigite or greigite–smythite mineralization can be compared with rocks of a genetically analogous type but partly or completely carbonified under natural conditions.

2. Oil shales of the Krkonoše (Riesengebirge) Piedmont basin, northern Bohemia

Samples were collected from the Rudník horizon, which occurs in the lower part of the Vrchlabí Formation in the Lower Permian Krkonoše Piedmont basin. The filling of this basin is characterized by cyclic sedimentation of red-bed type. Volcanic activity was recorded in different stratigraphic levels. During sedimentation the climate was semi-arid with humidity fluctuations. The Autunian age of the Rudník horizon is testified to by the occurrence of a specific macroflora, with several species of the genus *Callipteris brog*

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(Havlena, 1957). The Vrchlabí Formation was defined on the basis of regional lithostratigraphic division (Tásler et al., 1981).

The Rudník horizon is developed in the northern part of the Krkonoše Piedmont basin stretching in an east–west direction; towards the south the horizon thins away in psephitic–psammitic lithofacies. In its optimum development it attains a thickness of 60 m, in the outcropping part it can be observed along a length of about 25 km. Limnic sediments of the Rudník horizon are represented by green–grey and grey siltstones and claystones with subordinate beds of fine-grained sandstones. Oil shales are conspicuous in these actually black clayey shales as classified by Kukal (1985), i.e. claystones enriched with micro-organic matter. Organic matter, amounting to as much as 30% is composed mainly of kerogene; on average the C-bit. + C-hum. content amounts to 4%. In the course of epigenesis, the organic matter was remobilized, and at places laminar accumulation of the bituminous component was as high as 60% (Šimánek, 1986). Concretions, laminae to layers of carbonates, are formed by calcite and dolomite with variable contents of iron and manganese components; the recrystallization is frequent. Dispersed fine-grained analcite is usually remobilized in coarser grained laminae, nests and epigenetic veinlets with barite and anhydrite. Prior to palaeomagnetic research, these rocks were found to contain iron and copper sulphides (pyrite, chalcosite, marcasite, chalcopyrite and bornite); more rarely galena and sphalerite have been identified. Locally, phosphate pellets and phosphatized remains appear.

3. Palaeomagnetic data of oil shales

A total of 50 oriented samples from three sites of two localities about 20 km apart were studied. The dependence of normalized remanent magne-

TABLE 1

Thermal demagnetization of Autunian oil shales (Krkonoše Piedmont basin)

(a) Site 1A, 5781 A–5791 A ($n = 22$); locality, Vrchlabí

	Demagnetization							
	NS	100 °C	200 °C	250 °C	300 °C	330 °C	360 °C	390 °C
Declination	212.6°	204.6°	203.4°	203.4°	203.4°	203.3°	203.0°	203.0°
Inclination	10.2°	−0.6°	−7.2°	−8.1°	−8.0°	−9.0°	−8.6°	−6.7°
k	39.3	108.7	113.8	112.4	112.8	113.4	102.0	62.7
α	5.0°	3.0°	2.9°	2.9°	2.9°	2.9°	3.1°	4.0°

(b) Site 1B, 5792 A–5801 A ($n = 18$); locality, Vrchlabí

	Demagnetization							
	NS	100 °C	200 °C	250 °C	300 °C	330 °C	360 °C	390 °C
Declination	215.4°	205.4°	203.1°	203.3°	203.5°	205.3°	205.8°	[a]
Inclination	9.7°	−1.2°	−11.0°	−11.7°	−12.6°	−12.5°	−12.6°	[a]
k	50.5	317.7	522.6	266.4	287.3	100.0	78.1	[a]
α	4.9°	1.9°	1.5°	2.1°	2.0°	3.5°	3.9°	[a]

(c) Site 2, 5806 A–5811 A ($n = 10$); locality, Košťálov

	Demagnetization							
	NS	100 °C	200 °C	250 °C	300 °C	330 °C	360 °C	390 °C
Declination	238.3°	204.5°	197.2°	195.9°	195.6°	197.5°	195.8°	[a]
Inclination	54.8°	19.4°	0.4°	−2.1°	−3.0°	−6.0°	−5.6°	[a]
k	7.9	27.5	98.5	91.1	94.7	96.5	92.5	[a]
α	18.4°	9.4°	4.9°	5.1°	5.0°	4.9°	5.1°	[a]

[a] Samples partly disintegrated.

tization M_t/M_0 on temperature was investigated, where M_t is the intensity after stepwise demagnetization in zero field at temperature t, M_0 being the initial intensity. We also studied the dependence of volume magnetic susceptibility χ_t/χ_0 on heating. To measure remanence, we used JR-4 and JR-5 spinner magnetometers and a KLY-2 kappa-bridge (Jelínek, 1966, 1973) to measure magnetic susceptibility. Under thermal demagnetization, MAVACS apparatus was employed (Příhoda et al., 1989). For all the samples Zijderveld diagrams and stereographic projections of remanence directions in the course of thermal demagnetization were plotted. The graphs of thermal demagnetization of oil shale samples are represented by a typical example in Fig. 1. Phase changes occur at a temperature of 420 ° C, accompanied by an increase in magnetic susceptibility values, in some samples by as much as 80 times compared with values in the natural state. From the figure it ensues that the B component of remanence can easily be derived up to a temperature of 420 ° C; at higher temperatures deviations set in as a result of mineralogical changes, and are attended by an enormous increase in susceptibility. A component corresponds to viscous magnetization. The samples of the red-bed type collected from the same profile were stable up to the highest temperatures, making it possible to determine the Curie temperature of haematite as the remanence carrier.

Table 1 summarizes the results of Fisher's (1953) statistics of remanence directions during thermal demagnetization. The directions of B components can best be characterized after demagnetization in the temperature interval of 200–330 ° C. The results of samples from site 1A are presented in Fig. 2, the summary results for sites 1A, 1B and 2 in Fig. 3. The directions of B components of all the samples under study exhibit a strikingly close grouping, the directions coinciding with the palaeomagnetic directions of Autunian rocks of the Bohemian Massif (see Krs, 1968).

Table 2 summarizes palaeomagnetic directions for Autunian oil shales for sites 1A, 1B and 2 and the respective positions of virtual poles. The table also gives the mean palaeomagnetic directions

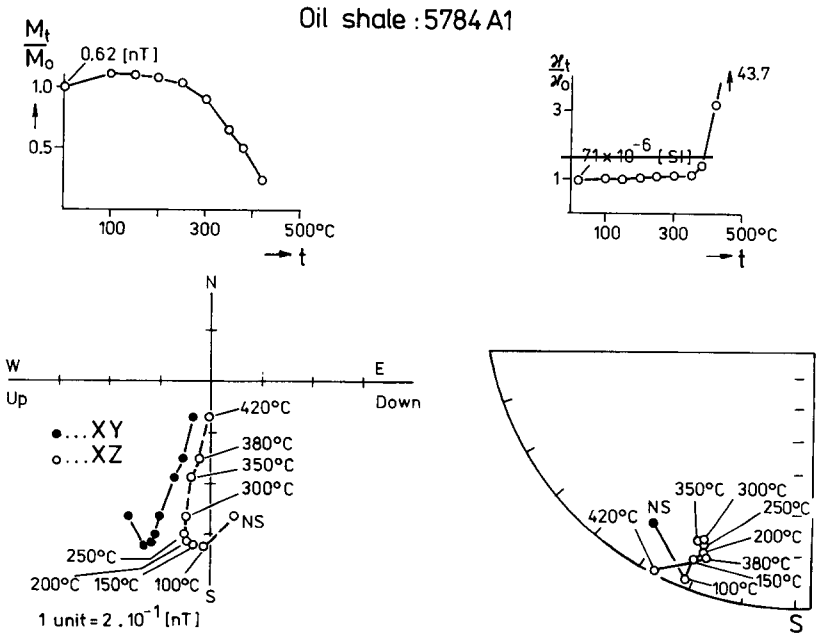


Fig. 1. Thermal demagnetization of Autunian oil shale, Vrchlabí, road cutting. Values of remanent magnetic polarization are given in nanotesla.

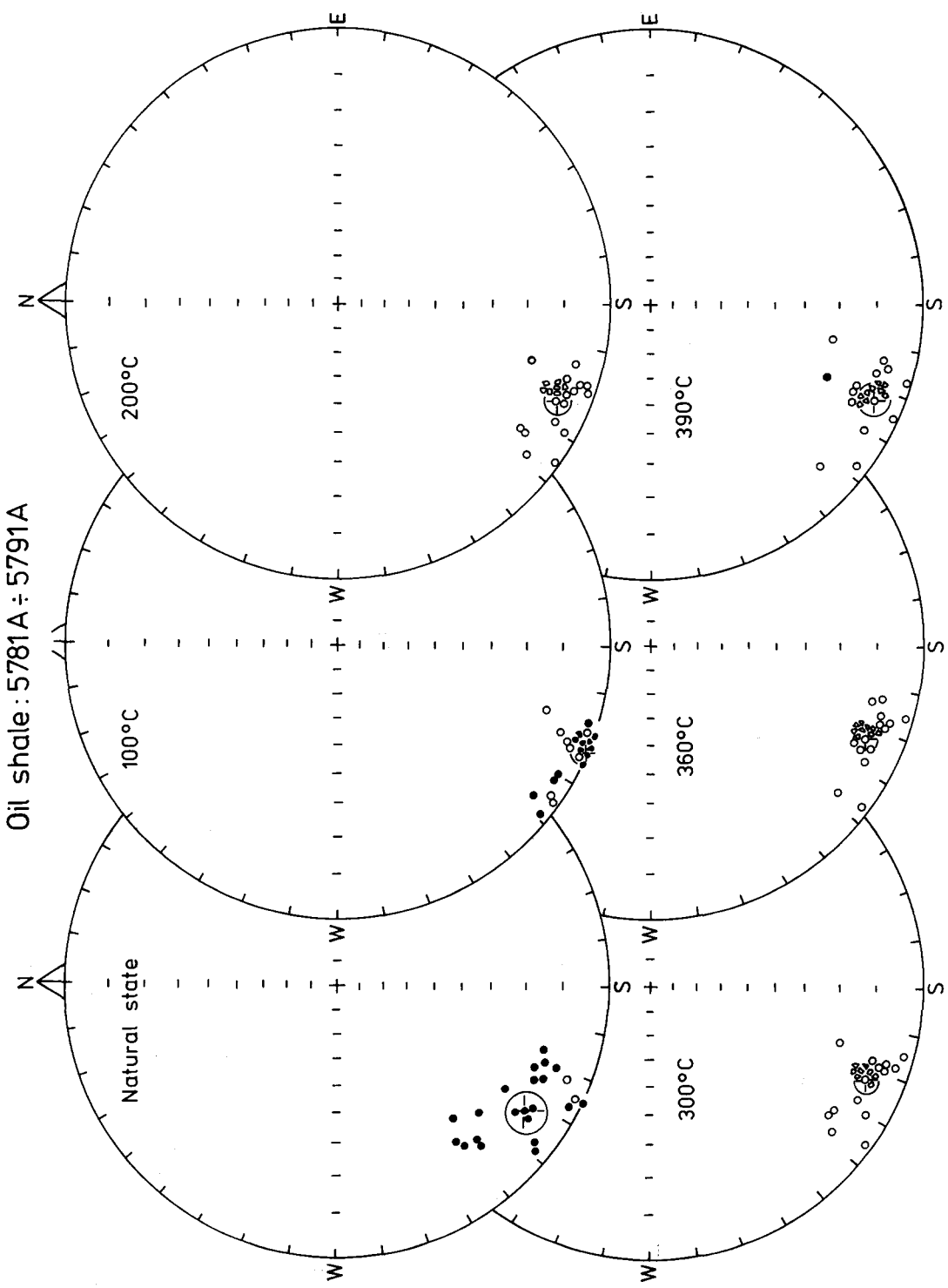


Fig. 2. Directions of remanence of samples of oil shale in natural state and of samples thermally demagnetized at temperatures of 100, 200, 300, 360 and 390 °C using the MAVACS apparatus, Vrchlabí.

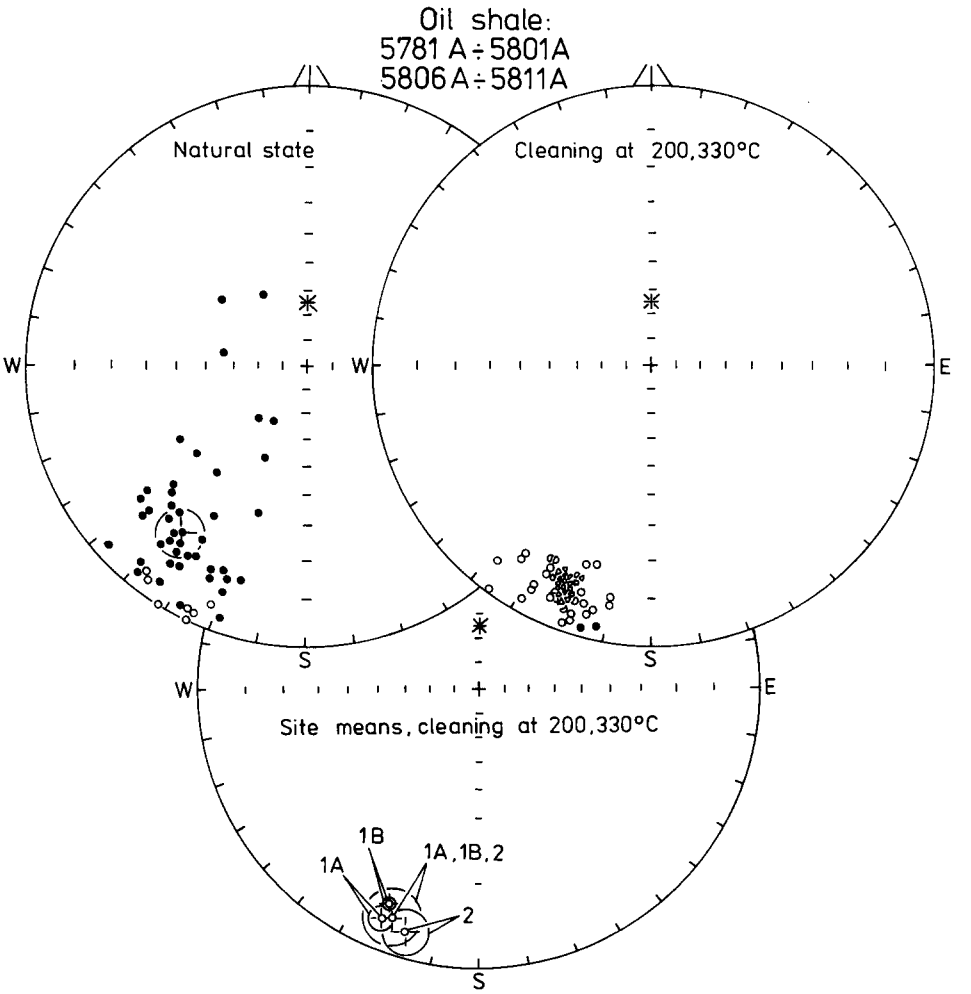


Fig. 3. Directions of remanence of samples in natural state and palaeomagnetic directions of samples of oil shales derived by progressive thermal demagnetization using the MAVACS apparatus, Krkonoše Piedmont basin.

TABLE 2

Autunian oil shale, Krkonoše Piedmont basin, Bohemian Massif

Locality	Site	Geographical coordinates		Palaeomagnetic directions		α	k	n	Pole positions		Ovals of confidence	
		φ	λ	D	I				φ_p	λ_p	d_m	d_p
Vrchlabí	1 A	50.624° N	15.599° E	203.4°	−7.2°	2.9°	113.8	22	39.04° N	164.92° E	2.9°	1.5°
Vrchlabí	1 B	50.624° N	15.599° E	203.1°	−11.0°	1.5°	522.6	18	40.98° N	164.51° E	1.5°	0.8°
Košťálov "Kovářův mlýn"	2	50.574° N	15.398° E	197.5°	−6.0°	4.9°	96.5	10	40.20° N	172.25° E	4.9°	2.5°
Vrchlabí	1A+ 1B	50.624° N	15.599° E	203.2°	−8.9°	1.8°	163.2	40	39.92° N	164.80° E	1.8°	0.9°
Vrchlabí+ Košťálov	1A, B +2	50.599° N	15.499° E	202.1°	−8.3°	1.8°	128.4	50	40.00° N	166.21° E	1.8°	0.9°

Mean pole position calculated from three virtual pole positions (1A, 1B, 2): $\varphi_p = 40.12°$ N; $\lambda_p = 167.22°$ E; $\alpha = 5.3°$.

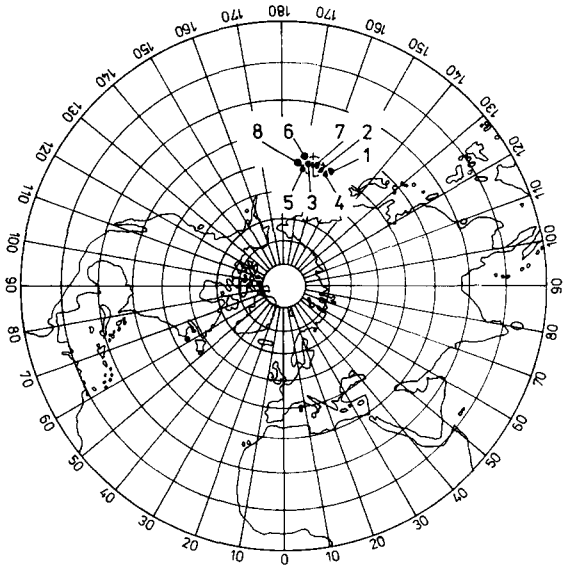


Fig. 4. Lower Permian palaeomagnetic pole positions so far derived for the Bohemian Massif.

computed from all samples (n = 50, strata means). Figure 4 makes it evident that the pole positions of Autunian oil shales fall well in the Autunian pole positions in the Bohemian Massif computed from palaeomagnetic directions of rocks of the red-bed type.

Figure 5 shows the mean palaeomagnetic directions of Lower Permian and Upper Carboniferous rocks representing the so-called strata means for various black-coal basins and furrows filled with sediments of the red-bed type, which were derived earlier (Krs, 1968). These results are supplemented with the data derived in Lower Permian oil shales.

The coincidence of palaeomagnetic directions of oil shales with the directions derived in rocks of the red-bed type in the Krkonoše Piedmont basin is striking. For the sake of comparison, Fig. 5 shows also preliminarily derived palaeomagnetic directions of the lower Carboniferous rocks in the Upper Silesian black-coal basin. To give some evidence of the reliability of these preliminary palaeomagnetic data see the results of Fisher's (1953) statistics of remanence directions during thermal demagnetization. Table 3 summarizes results for the locality Krásné pole quarry, and for the locality Vondruškův lom quarry, near the village of Budišovice. Figure 6 demonstrates a typical Zijderveld diagram and normalized graphs for remanence and magnetic susceptibility. Palaeomagnetic investigations into rocks of the Ostrava–Karviná region are continuing.

4. Major results

The object of this brief communication was to demonstrate that oil shales from the Krkonoše

TABLE 3

Thermal demagnetization of coal-bearing Lower Carboniferous shales (Upper Silesian basin, Ostrava–Karviná region)

(a) Sample Nos: 5581 A–5604 A; locality, Krásné pole quarry

	Demagnetization						
	NS	100 °C	150 °C	200 °C	250 °C	300 °C	350 °C
Declination	346.7°	214.2°	216.1°	216.8°	215.4°	214.3°	206.4°
Inclination	79.2°	31.3°	13.2°	13.6°	8.0°	3.3°	−1.0°
k	3.2	5.2	5.2	4.3	3.8	6.1	6.1
α	19.6°	14.2°	14.1°	15.8°	17.4°	12.9°	12.9°

(b) Sample Nos: 5605 A–5622 A (n = 28); locality, Vondruškův lom quarry

	Demagnetization							
	NS	100 °C	150 °C	200 °C	250 °C	300 °C	350 °C	400 °C
Declination	210.4°	205.4°	203.4°	200.8°	199.4°	201.8°	199.1°	200.7°
Inclination	−0.8°	−6.0°	−10.5°	−9.8°	−11.5°	−12.6°	−13.1°	−15.0°
k	40.7	43.0	54.2	45.2	35.8	29.8	13.9	10.0
α	4.3°	4.2°	3.7°	4.1°	4.6°	5.1°	7.6°	9.1°

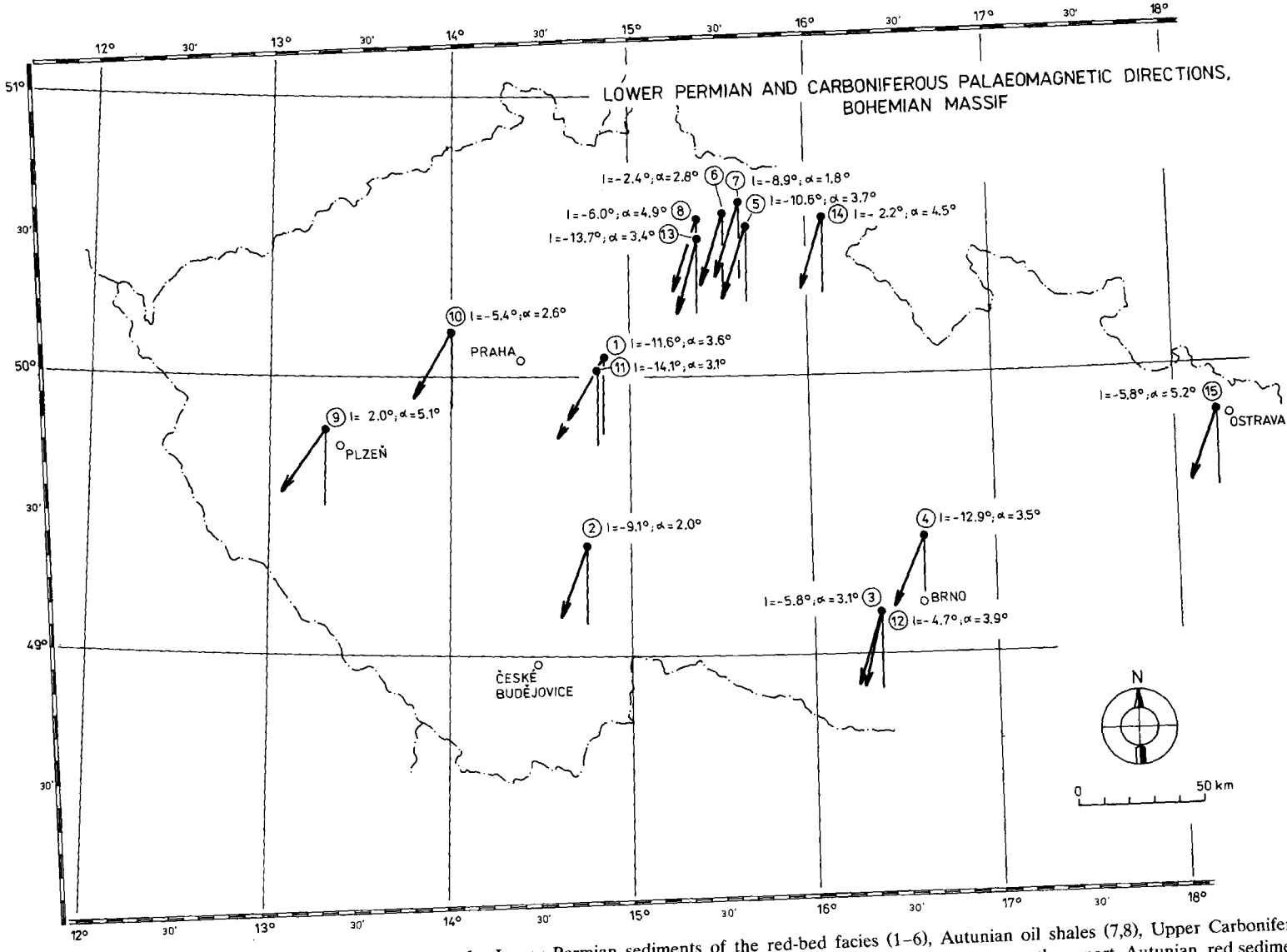


Fig. 5. Bohemian Massif: palaeomagnetic directions for Lower Permian sediments of the red-bed facies (1–6), Autunian oil shales (7,8), Upper Carboniferous sediments of the red-bed facies (9–14) and Lower Carboniferous coal-bearing shales (15). Lower Permian: 1, Blanice furrow, northern part, Autunian, red sediments, $n = 40$; 4, Boskovice furrow, central part, Autunian, red sediments, $n = 43$; 2, Blanice furrow, southern part, Autunian, red sediments, $n = 34$; 3, Boskovice furrow, southern part, Autunian, red sediments, $n = 116$; 6, Krkonoše Piedmont basin, Lower 'Rotliegendes', red sediments, $n = 45$; 5, Krkonoše Piedmont basin, Upper and Middle 'Rotliegendes', red sediments, $n = 89$; 7, Krkonoše Piedmont basin, Autunian, oil shales, $n = 40$; 8, Krkonoše Piedmont basin, Autunian, oil shales, $n = 10$. Upper Carboniferous: 9, Plzeň basin, Westphalian–Stephanian, red sediments, $n = 65$; 10, Kladno–Rakovník basin, Westphalian C–Stephanian, red sediments, $n = 73$; 11, Blanice furrow, Stephanian, red sediments, $n = 25$; 12, Boskovice furrow, Stephanian, red sediments, $n = 40$; 13, Krkonoše Piedmont basin, Stephanian, red sediments, $n = 57$; 14, Inner Sudetic basin, Westphalian B–Stephanian, red sediments, tuffs, $n = 88$. Lower Carboniferous: 15, Ostrava–Karviná region, Upper Silesian basin, Lower Carboniferous, coal-bearing shales, $n = 53$.

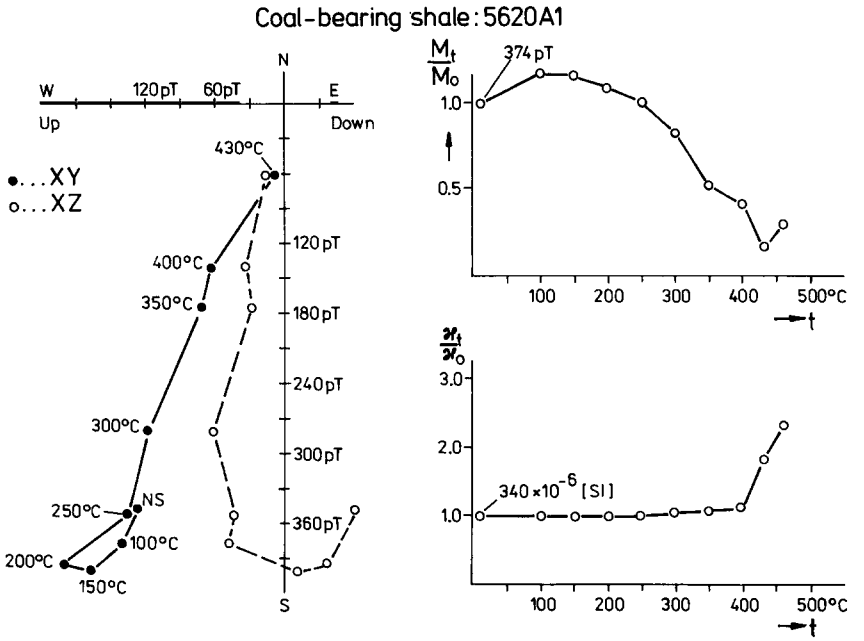


Fig. 6. Thermal demagnetization of Lower Carboniferous coal-bearing shale, Vondruškův lom quarry, Ostrava–Karviná region, Upper Silesian black-coal basin. Remanent magnetization directions are given in the stereographic projection; full (open) circle denotes projection into the lower (upper) hemisphere. The mean directions were computed with the use of Fisher's (1953) statistics; the mean direction is denoted by a circle with a dagger, around which the confidence circle is circumscribed for a probability level of 95%.

Piedmont basin were found suitable for palaeomagnetic research. Of interest is the high coincidence of palaeomagnetic data on rocks of different genesis in the territory of the Bohemian Massif, which points to tectonic stability of the Bohemian Massif blocks in the period following the Lower Permian. The results obtained from oil shales are in complete agreement with these data. For rocks of the Upper Carboniferous the scatter of palaeomagnetic data is greater, but even preliminary results obtained in the Ostrava–Karviná region, Upper Silesian basin, indicate a palaeogeographic affinity of the basin to the blocks of the Bohemian Massif. Data presented in the present communication indicate that rocks containing micro-organic matter at different stages of carbonification, such as bituminous rocks, black shales, oil shales, coal-bearing rocks etc., may be suitable for palaeomagnetic research.

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